

UBUNTU SERVER INSTALLATION GUIDE

Week 3 - Enterprise Server Deployment

ITEP 414 - System Administration and Maintenance

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Project Overview

This guide documents the deployment of an Ubuntu Server virtual machine for ABC Startup Solutions. The target server will provide a foundation for future file sharing, remote administration, database hosting, web hosting, and internal services.

Objectives

- Install Ubuntu Server in a virtual machine.
- Configure hostname, administrative user, storage, networking, and OpenSSH.
- Verify server identity, connectivity, updates, and SSH service.
- Document the deployment so another administrator can reproduce it.

Software Requirements

Software / Image	Purpose
Oracle VirtualBox or VMware Workstation	Virtual machine platform
Ubuntu Server LTS ISO	Linux server installation media
Windows Server Evaluation ISO	Bring-home comparison activity

Virtual Machine Specifications

Component	Required Configuration
Name	Ubuntu-Server-Week03
RAM	4 GB minimum
CPU	2 virtual processors
Storage	40 GB VDI/VMDK
Network	NAT, or Bridged if instructed
Optical Drive	Ubuntu Server LTS ISO

Step-by-Step Ubuntu Server Installation

1. Create the Virtual Machine

Create a VM named Ubuntu-Server-Week03 with at least 4 GB RAM, 2 virtual CPUs, a 40 GB virtual disk, and NAT networking. Attach the Ubuntu Server LTS ISO.

2. Start the Installer

Boot the VM from the ISO and select English.

3. Keyboard Layout

Select the appropriate layout, normally English (US).

4. Network Configuration

Accept DHCP unless your instructor requires a static address. Record the assigned IP address.

5. Storage

Choose Guided - Use Entire Disk. Record the disk size, partition layout, and filesystem shown by the installer before confirming.

6. Profile / Hostname

Set the server hostname to server01. Create a non-root administrative account using your own name and username, with a strong password.

7. SSH

Enable Install OpenSSH Server so the server can be administered remotely.

8. Additional Packages

Do not select additional packages unless instructed.

9. Complete Installation

Finish installation, reboot the VM, and remove/eject the installation ISO when prompted.

Configuration Details to Record

Item	Your Value
Hostname	server01
Username	_____
IP Address	_____
Disk Size	40 GB
Partition / Filesystem	_____
Network Mode	NAT / Bridged
SSH	Enabled

Server Verification

Login

Log in using the non-root administrative account.

[INSERT YOUR SCREENSHOT HERE: Login]

Recommended filename: screenshots/01-ubuntu-login.png

Hostname

Run: hostname. Expected hostname: server01.

[INSERT YOUR SCREENSHOT HERE: Hostname]

Recommended filename: screenshots/02-hostname.png

IP Address

Run: ip addr. Capture the active interface and assigned IP.

[INSERT YOUR SCREENSHOT HERE: IP Address]

Recommended filename: screenshots/03-ip-address.png

Internet Connectivity

Run: ping -c 4 google.com. Capture the successful replies.

[INSERT YOUR SCREENSHOT HERE: Internet Connectivity]

Recommended filename: screenshots/04-ping-google.png

System Update

Run: `sudo apt update`, then `sudo apt upgrade -y`. Capture successful update/upgrade output.

[INSERT YOUR SCREENSHOT HERE: System Update]

Recommended filename: screenshots/05-apt-update-upgrade.png

SSH Service

Run: `systemctl status ssh`. It should show active (running).

[INSERT YOUR SCREENSHOT HERE: SSH Service]

Recommended filename: screenshots/06-ssh-status.png

Problems Encountered

Record only problems you actually encounter during your installation. Do not invent issues. For each problem, include the symptom, likely cause, action taken, and final result.

Lessons Learned

The deployment demonstrates that server installation must be followed by verification and documentation. Hostname, addressing, updates, connectivity, and remote administration are basic checks that establish whether a newly installed server is ready for further configuration.

References

Week 3 Portfolio Project - ITEP 414 module.

Ubuntu Server documentation - <https://help.ubuntu.com/>